Profiling Ethics Orientation through Play

Carlos Pereira Santos[ab]*, Vassilis-Javed Khan[b] and Panos Markopoulos[b]

[a] *Academy for Digital Entertainment, Breda University of Applied Sciences, Breda, , The Netherlands;* [b] *Department of Industrial Design, Eindhoven University of Technology, Eindhoven, The Netherlands*

* santos.c@nhtv.nl or c.a.pereira.santos@tue.nl

P.O. Box 3917

4800 DX Breda

The Netherlands

T +31 76 533 22 03

Profiling Ethics Orientation through Play

Research studies and recruitment processes often rely on psychometric instruments to profile respondents with regards to their ethical orientation. Completing such questionnaires can be tedious and is prone to self-presentation bias. Noting how video games often expose players to complex plots, filled with dilemmas and morally dubious options, the opportunity emerges to evaluate player's moral orientation by analysing their in-game behaviour. In order to explore the feasibility of such an approach we examine how users' moral judgment correlates with choices they make in non-linear narratives, frequently present in video games. An interactive narrative presenting several moral dilemmas was created. An initial user study (N=80) revealed only weak correlations between the users' choices and their ethical inclinations in all ethical scales. However, by training a genetic algorithm on this data set to quantify the influence of each branch on recognizing moral inclination we found a strong positive correlation between choice behaviour and self-reported ethical inclinations on a second independent group of participants (N=20). The contribution of this work is to demonstrate how genetic algorithms can be applied in interactive stories to profile users' ethical stance.

Keywords: games user research; moral judgment; ethics; game design; genetic algorithm.

Word count: 6950

Introduction

Following the financial crisis of 2008 there has been a growing concern surrounding news reports regarding unethical practices in business. Unethical behaviours like stealing, lying or fraud are affecting adversely the economy but also society in general (Imanishi-kari, and Hill 2006; KPMG 2013; Cohen et al. 2010; Soltani 2014). In a recent analysis of 2,410 cases of occupational fraud reported between 2014 and 2015 it is argued that the studied organizations lost about 5% of their annual revenue as a result of fraud while the total losses resulting from unethical behaviours exceeded $6.3 billion,

with an average loss per case of $2.7 million (Association of Certified Faud Examiners 2016). Growing public awareness of unethical behaviour in business has prompted calls for explicit actions to instil ethical values in corporations (Cherrington and Cherrington 1992), which makes methods for assessing the moral judgement of employees very relevant.

Moral judgment is the thinking process that determines if an idea, entity or action is right or wrong; but also how it fares in terms of other related concepts such as good, bad, unconventional, bizarre, ethical or irrational, to name a few. The process an individual takes to determine "right" or "wrong" is directly dependent upon the individual's cultural setting, context and the perceived consequences an action might have (McDevitt, Giapponi, and Tromley 2007). People often find themselves in a situation where a difficult choice has to be made between two or more undesirable alternatives, otherwise known as a dilemma. People face ethical dilemmas in their professional capacity, e.g., end of life judgments for medical doctors (Lloyd 2004), in major choices pertaining to family and social life, but even in mundane daily activities where people break social norms or laws; e.g., cutting in line to make it for lunch, or speeding while driving to catch a flight.

Different ethical theories provide guidance for resolving moral judgments based on diverse and sometimes conflicting values. Representative theories include: Moral equity, Contractualism, Relativism, Egoism and Utilitarianism (McDevitt, Giapponi, and Tromley 2007). The conflict between these values underlies classic dilemmas (e.g., Heinz's dilemma or the trolley problem, (Kohlberg 1981)), and is manifested in numerous entertainment products like movies, music, books, and recently even games.

Games as profiling tools

Games are a medium promoting very extensive and diverse player interactions, which

encourages game companies to use telemetry to monitor player's in-game behaviour. Analysing such data can provide clear and detailed information about what players do within the game, which can serve marketing purposes or the development of more engaging games. In the last few years this practice has rapidly expanded and matured across the gaming industry, which is increasingly recognizing the value of understanding player behaviour (El-Nasr, Drachen, and Canossa 2013).

The term "Game Metric" is defined as data-driven information that is collected with games. Game metrics are interpretable measures of in-game interactions by the players (Yannakakis et al. 2013), e.g., at what times do people play a game, how much time or money they spend. Game metrics often serve predictive business analyses pertaining to topics such as the expected sales figures or the expected number of players. Other examples of game metrics concern in-game player behaviour that can be valuable during game design, authoring and content creation, e.g., in-game position, frequency of used items, accuracy and skill set, which allow game designers to know more about the players, their behaviour, proficiency and experience.

Player Modelling (Yannakakis et al. 2013) pertains to developing means to measure, interpret and represent player behaviour (Nacke et al. 2009). Recently, there has been an effort to assess personality profiles based on in-game behaviour, like the Big Five Personality Traits (Yee et al. 2011; Lankveld, Schreurs, and Spronck 2009; Lankveld et al. 2011), HEXACO (Worth and Book 2014), Need for Cognition (Pereira Santos, Khan, and Markopoulos 2016) and Self-esteem (Pereira Santos et al. 2017). In these studies videogames are seen as a sandbox that encourages role-play and that allows players to make choices and take actions while avoiding their real-life consequences, in a way that reflects their individual personality.

This research aims to contribute to Player Modelling literature by showing how in-game choices relate to a player's moral judgment. Specifically, this study addresses the following research questions:

- Is players' moral judgment related to their in-game behaviour?
- How can an interactive narrative be designed to allow implicit profiling of players' ethical orientation?

The contribution of this paper is twofold. First, it pertains to application of machine learning algorithms to relate users' choices in interactive stories with their moral judgment. We show that choices players make during interactive narratives can serve as an implicit measure of one's ethical judgment. This can be useful as an alternative to psychometric questionnaires, with some advantages over those, such as being more fun and engaging, and potentially leaving less room for self-presentation bias when compared to explicit self-report.

Multi-Dimensional Ethics Scale

Most profiling ethics methods require participants to self-report opinions of specific situations. Perhaps the most used psychometric instrument for understanding ethical norms is the Multidimensional Ethics Scale (MES) which was originally developed by Reidenbach and Robin (Randall and Gibson 1990) and then further refined by Reidenbach et al. (R. E. Reidenbach and Robin 1990).

MES represents the evaluative criteria that individuals use in making a moral judgment based on five independent ethical norms: moral equity, deontology, relativism, egoism and utilitarianism. Scores along these scales represent a 'moral compass' of respondents. Since its publication, numerous studies have used MES to evaluate all five or a subset of the five ethical norms, demonstrating a strong degree of

validity and a high degree of reliability. MES has been applied in a wide range of contexts, including evaluating marketing ethics (R. Eric Reidenbach, Robin, and Dawson 1991), business ethics (Gupta 2010) and questionable Internet practices (Williamson et al. 2011).

In the MES questionnaire a participant evaluates a set of vignettes according to several semantic scales, i.e. a short story is presented to the participant that explains the setting of a dilemma or an ethical issue and followed by an action explaining how the situation was resolved. The participant is then requested to evaluate the action by classifying (in a Likert scale) multiple semantic statements that are associated with the different ethical norms, like just-unjust, fair-unfair, does-doesn't violate an unspoken promise, etc. (Randall and Gibson 1990) (see Figure 1).

Ethical Norms

A complete overview of all ethical norms associated with Moral Philosophy is outside the scope of this paper; readers can find more information in relevant textbooks (McDevitt, Giapponi, and Tromley 2007). It is quite common in research to address only a subset of specific and distinct ethical theories rather than the complete MES inventory (Buchan 2014; Jung 2009; Flory et al. 1992; Humphreys et al. 1993). Accordingly and to keep the size of the study manageable, we demonstrate the feasibility of our approach by considering only three out of the five ethical norms as including all five norms would increase considerably the writing effort and the logistics of experiments. Specifically, we addressed the following norms: Contractualism, Utilitarianism and Moral Equity. A brief discussion of all ethical norms is provided as background information for this study.

Moral Equity.

Much of the most influential concepts of moral equity (also known as Justice) derive from the writings of Aristotle. The Principle of Formal Justice states: "Treat similar cases similarly and different cases differently in the proportion to their difference(s)" (Carr 1981). It is considered formal because it specifies the form that matters in justice but still does not specify what counts as alike and unalike, leaving that for a substantive principle of justice (Carr 1981). The most important concept in this theory is related to the equal distribution of value and to the develop rules or procedures that result in fair or just outcomes. Moral equity can be subdivided into procedural and distributive justice.

Procedural justice relates to the fairness in processes, and it is divided into: *i) pure*, if the rules guarantee an objectively just outcome in every occurrence, *ii) perfect*, if the rules provide an individual with a fair result in every case, and *iii) imperfect* procedural justice, if the rules represent the best attempt to produce fair results, but sometimes the outcomes are unjust.

Distributive justice is a contemporary ethical theory focused on the uniform distribution of values and/or goods (Rawls 1999). An action is ethical if it provides equal wealth, rewards, opportunities, obligations and/or burdens by all involved members, unless there is a morally significant and justifiable difference between them.

Contractualism

Contractualism also known as Deontology, stems from the Greek word deon (δέον), which means "duty", pertaining directly to an individual's duty towards others. In other words, it is the duty of an individual to act according to genuine and legitimate claims of others, even if not enforced by any authority (Scanlon 1982). This ethical theory

defines, for example, the sense of responsibility to pay our debts or care for family. Actions are the main drivers of duty ethics, which teaches that some acts are right or wrong, and individuals have a duty to act accordingly, regardless of the good or bad consequences that may be produced.

Emmanuel Kant is the father of the "Categorical Imperative", which uses the notion of maxims to define moral acts. Maxims are universal rules that define what ethical actions are. Simple examples are: it is wrong to kill innocent people, it is wrong to steal, or it is right to keep promises (Kant 2000). As an extreme though illustrative example, Kant argued that it would be wrong to tell a lie in order to save a friend from a murderer. This ethical theory is linked with the individual's rights, by enforcing duties and codes of honour for individuals to follow. A general understanding of maxims comes through political, social or religious grounds like constitutions, military codes or sacred texts.

Utilitarianism

Utilitarianism states that individuals should act to produce the greatest possible good for all of society. It requires the decision maker to consider all possible outcomes of their actions/inactions and weigh which results in the best ratio of good versus bad outcomes for society (Mulgan 2007). Effectiveness and productivity are also very much related to this theory, since a less efficient action is likely to produce less utility and would therefore be qualified as less ethical. Also, concepts like democracy where the opinion of the majority rules, demand and supply or capitalism are linked to utilitarianism theories (Scanlon 1982; Mulgan 2007).

An important notion to understand is the fact that according to utilitarianism, actions that may cause harm to individuals or minorities can be defined as ethical if they can be compensated with a minor gain to a majority. An extreme example would be to

refuse medical care to someone during rush hour because it will create a traffic jam for thousands of people. In addition, the definition of greater good can be broken down into multiple parts like: social, economic and ecological. Historically those aspects often clash, and it is in practice difficult to define and quantify ethical action based on imperfect information and the inability to measure or compare outcomes (Scanlon 1982).

Ethical Relativism

Relativists believe that ethical values are dependent on one's culture, feelings, or religion. Thus a decision may be deemed ethical in one's own culture, while the same behaviour under the same circumstances may be judged unethical in another culture. An individual has to be careful and distinguish what people believe is right or wrong, and there can be no objective basis for a moral judgment as every culture devises its own set of moral standards, and therefore, no universal ethical rules exist that apply to all (Newman and McFarland 1990).

Anthropologists have reported diverse and contradictory ethical values in different cultures, which is the main evidence and the justification for relativism. There is some criticism over relativism, pertaining to it not serving the main task of ethics: the preservation of conditions that allow people to pursue a happy and stable life (Hoffman, Frederick, and Schwartz 2014). Several managers cite relativism as a justification for alleged unethical behaviour, especially in multi-cultural and international situations (Hoffman, Frederick, and Schwartz 2014).

Egoism

Egoism is a teleological (consequentialist) theory that states decisions should reflect solely on one's best long-term interests, i.e., an ethical action should be performed if it

produces a greater ratio of good versus evil for the individual in the long run. In short, it means actions that result in the greatest good for the individual. Some variations of this theory focus on a short-term hedonism, stating that everyone is psychologically programmed to behave only in their own self-interest (Rand 1964).

Modern philosophers consider egoism seriously since the personal long-term interests create a suitable context, which motivates individuals to help others, as well as help to define and follow the rules of society. There are some critical positions against egoism, namely that it disregards dogmatic principles, like what most people commonly agree is right and wrong. In addition, and similar to relativism it does not serve the purpose of ethical philosophy (Longenecker, McKinney, and Moore 1988).

Games and Ethics

Earlier work looking directly into games and ethics is sparse. Zagal (Zagal 2009) reviews several video games notable for their moral dilemmas and discusses ethical settings that are presented to players that challenge ethical norms. Zagal states that although games portray ethical codes similar to reality, it is possible that a game might define its own cultural settings and encode a new ethical system which the player is required to learn and embrace in order to succeed. Zagal distinguishes the following categories of ethical perspectives on games: Artefact (e.g. narrative), Business (e.g. development and production), Play (e.g. sportsmanship and cheating) and Gameplay (e.g. in-game behaviour actions). Zagal concludes that moral judgment can be used to elicit and create emotionally meaningful experiences but the full potential for ethical reasoning and reflection in games has not yet been explored.

Several scholars reflect on moral disengagement in video games (Sicart 2009; Hartmann and Vorderer 2010; Gabbiadini et al. 2014). While such critique and concerns are well founded, it is also important to note that a deeper understanding on how a

player's moral judgment relates to in-game choices is still missing, while it can precisely help measure the influence of games on players' ethics.

An attempt to profile players is reported in (Ewell et al. 2016), where a specific game mechanic (avatar building) is used to predict moral deviation. This work suggests that avatars represent not only their players' psychological traits, but also correlate with the moral disengagement scale which assesses the propensity to overlook or ignore one's own rules of moral conduct. Compared to that work, we are particularly interested in measuring players' alignment with ethical norms rather than just their tendency to apply them in their in-game behaviour.

Method

We set out to answer two research questions: i) "Is players' moral judgment related to their in-game behaviour?" and, ii) "How can an interactive narrative be designed to allow the implicit profiling of players' ethical orientation?" For this, we adapted a branching narrative and examined players' choices within the story and how their choices compare with their responses to the Multidimensional Ethics Scale (MES).

We compared the MES scores for 80 participants with the choices they made within the branching narrative by calculating the correlation between the two scales.

We purposely used just a branching narrative, stripped down of any other known game mechanics. With this setup we could prevent strong influences like character self-representation or rewards that might affect the players' drive towards a certain branch of the narrative and in that way bias the study. Branching narratives are considered as non-linear gameplay, which emphasizes on the player's choices as a central part of the design of the game. Using Crawford's game definition and taxonomy (Crawford 2008), the branching narrative is classified as an entertainment interactive challenge or puzzle (see Figure 2).

Branching Narrative

The branching narrative that we developed was inspired by the movie Interstellar (2014)1. Our story is based on a sci-fi setting and places the reader in command of a spaceship where he or she must manage with limited resources and a small crew in a last-ditch attempt to find a solution to a plagued and threatened Earth. The story is written in English, capturing the main plot of the movie and featuring similar characters. The narrative is around 10.000 words (including all branches), and our data shows that native English speakers take 30.5 minutes on average to complete a single read through (SD=9.1). The story was split in multiple small segments (from 10 to 250 words each) to be presented separately to readers and which provide distinct narrative sections. At the end of a section a way forward is presented with one or more links to following sections (see Figure 3).

Singular links in a section provide a linear progression, while multiple links provide readers with branching choices in the narrative. A reader is asked to commit to a certain choice and is not allowed to revise his or her decision, i.e., when given multiple choices, a reader can only choose one specific path, and there is no option to go back and modify a choice once it has been made. Across the narrative there are 15 moments that all readers go through independently of their previous choices and where they need to make a choice (illustrated as numbered nodes in Figure 3). In these specific 15 moments there are always three choices presented to them, each based on one of three aforementioned ethical norms.

To support the data collection a web-based platform has been developed, which allowed participants to read the story online and make their choices. The order of

---

[1] www.interstellarmovie.net

choices available to players is randomly presented in order to avoid placement bias. In-game metrics record all players' decisions. For the purposes of this study only the choices in these 15 moments matter. An example of a page of the narrative on one of those 15 moments with three options is depicted in Figure 4.

Participants

To reach a more diverse participant group than the typical university participant pools (Behrend et al. 2011), a crowdsourcing service (prolific.ac) was used and a small monetary reward (£2,50) was provided for completing the study. Detailed participant demographic profiles were provided by the crowdsourcing service, and all participants were informed of the fact that data was being collected for research purposes and that all logged data was anonymous.

All our participants were native English speakers, and the average age of our participants was 35.6 (SD=11.4), with 54% being female, almost all were Caucasian, but the sample showed good diversity on religion and political affiliations. Even though most of the participants belong to the Occidental culture, it is not expected that this study would be influenced by cultural differences; the plot reflects a futuristic environment and a global scale problem, avoiding references to countries or strong cultural ties. In addition, the collected data is measured against a validated model (MES), which has being used in multiple target audiences with diverse religions and cultural backgrounds.

Process

A typical participant followed the following steps:

- Receive an invite to play an online game

- Open a URL to start the branching narrative
- Complete a demographic questionnaire
- Complete the MES questionnaire
- Read the branching narrative, and chose an option for each of the branches (Figure 4)
- Receive the results of the MES questionnaire and their choices in the branching narrative (Figure 5).

To avoid learning effects and self-presentation bias, we only considered participants' first attempt at the game. Furthermore, we excluded participants that had previously seen the movie that our story was based on; a specific question was included in our demographic questionnaire for that purpose. Our last exclusion parameter related to how long people took to read the story, all outliers were also excluded, i.e., players that read the story under 12 minutes and above 49 minutes.

Results

After applying the aforementioned exclusion criteria, we retained 55 participants (out of 80). For the 55, the data was more or less evenly distributed among the different ethical theories in both the MES scores and the followed branches in the narrative (Figure 6). We note a slight preference of our sample for utilitarianism and contractualism over moral equity; this pattern is noticeable both in the MES scores and in the branching narrative choices.

In the following paragraphs we explore the relation between the choices made in the branching narrative and the players' MES scores.

Raw scores

As a first approach we correlated all branch choices to the MES scores. As an example to clarify the calculations; player p starts by filling out the MES questionnaire, from which we obtain an independent score for each of the different ethical norms MESp,e where e is the ethical norm (i.e. Moral Equity, Contractualism and Utilitarianism). MESp,e ranges from 0 to 48; it is calculated by adding a score ranging from 0 to 6 for two different attributes per norm for each of the four vignettes.

While the player reads we record all choices p made throughout the 15 branches (Figure 3). Bp,n=e represents player p's ethical norm choice on branch n. After p completes reading the story, we calculate the branch score BSp,e (1), which indicates how frequently e has been selected.

$$BS_{p,e} = \frac{\sum_{n=1}^{15} \begin{cases} B_{p,n} = e & 1 \\ else & 0 \end{cases}}{15} \quad (1)$$

In this analysis we calculate Spearman's correlation coefficient independently for every ethical norm e by comparing all the pairs (MESp,e, BSp,e) for all players. This analysis does not provide any evidence of a correlation between the choices made in the story and the MES scores (Table 1).

Machine Learning

Since a direct correlation was not obvious we decided to apply a machine learning approach. The approach we followed uses a genetic algorithm commonly used in feature construction and optimization algorithms (Bies et al. 2006; Goldberg 1989) to identify how much each branch contributes to the scores for each of the ethical norms. Using the branch choices from the 80 participants we calculated a weight Wn,e for each branch n which optimized the inference of the ethical orientation e. With these weights we wanted to see if they could be used to correlate with other participants' ethical

orientation.

The genetic algorithm rules are straightforward: the weight range is between 0.0 and 2.0, and the sum of all the weights needs to be 15.0 (1.0 per branch). The algorithm uses elitist selection (best solutions are stored from multiple generations), and the fitness function is the Spearman's statistical correlation. We aim to maximize the correlation (r) while minimizing the significance (p).

The We (set of Wn,e for all branches) were generated by: *i)* randomly generating the full set of weights, *ii)* mutating a single weight by assigning a new random value for a single branch and *iii)* by cross-over where new sets are generated by cross-breeding existing sets. The algorithm computed multiple generations of We, and convergence was achieved after 8 to 10 generations.

$$BS_{p,e} = \frac{\sum_{n=1}^{15} \begin{cases} B_{p,n} = e & W_{n,e} \\ else & 0 \end{cases}}{15} \quad (2)$$

Validation with a new data set

We recruited a new cohort of participants (N=20) from Prolific.ac. These 20 participants showed similarities with our original group; their average age was 36.9 (SD=11.2) and 50% were female. They followed the same process and similarly we excluded three participants applying the same exclusion criteria as before. The difference with this group pertained solely to how we compute the branch score, by applying the Wn,e scores that were calculated with the data collected from the first group of participants.

The results (see

Tables

) show strong positive correlations between the new set of readers' narrative choices and their MES scores, allowing us to conclude that there is a relation between players' in-game choices and their moral judgment, when appropriate weights are applied.

Discussion

A strong narrative intertwined with gameplay often helps create compelling ethical dilemmas. Game designers put a lot of effort in creating an authentic interactive story and an engaging experience. On the other hand, it is also common for designers to use dilemmas within the narrative to create emotional bonds with the player's decisions and give players the feeling that they are actually in the starring role with meaningful consequences. The use of ethical theories within game narrative design is not entirely new, as some designers actively study them to apply them within game development (Sicart 2009). However, our study has taken a different approach to examine if and how the players' moral judgment is manifested in their gameplay. By uncovering the correlation between in-game player choices with their moral judgment we show *i)* that players' moral judgment is related to their in-game behaviour, and *ii)* how to design an interactive narrative support the implicit profiling of players' ethical orientation.

Profiling moral judgment through play

Our results support the hypothesis that player in-game choices are influenced by their moral judgments. Initial analysis of the raw data from all recorded choice behaviours did not reveal any correlation to their moral judgment scores obtained through MES. However, using a machine learning approach to weigh the contribution of each node of the interactive narrative, we could obtain a score that describes the

player's observed moral choice behaviour which strongly correlates to the MES score. With this work we contribute to the literature by showing how an interactive story can provide a measure that correlates to self-reported ethical judgments and how the application of a machine learning approach which designers of interactive narratives can apply to profile their players' moral judgments.

While this result was obtained in the context of an interactive narrative, we conjecture that it could be possible to compute a similar correlate to players' ethical judgments with any game (also of a different genre) that places the player in a position which he or she has to choose between two or more alternatives that correspond to different ethical choices as a way to profile players' moral judgment, like: survival, role-playing, construction, management, stealth, life simulation or war games. To do so successfully it is important to map actions and choices that are available to the player on ethical norms. Our results also show that not all branches are effective identifiers for ethical norms. In the next section we discuss in more detail these differences.

Designing effective Choice/Actions identifiers

A careful look at the narrative of high-weighted nodes shows that unambiguous contexts, referring to known terms that are associated with norms are better able to evoke choice behaviour that relates to players' ethical norms. In the example depicted in Figure 4 (B13), it is clear that the first option is linked with utilitarianism and represents consequences ("might lose"), while the second option is linked with contractualism ("mission"), and finally the third is linked with moral equity ("fair").

On the other hand, a complex introduction or less clear choice statements can lead to different moral interpretations. For example, in B5 readers have to decide in which planet to land first and a long story plot is revealed with multiple side effects. In addition the choices presented to the user may reflect different norms: *i)* 'What if he

needs our help?', *ii)* 'We should conserve fuel' and *iii)* 'We will do what the majority wants'. These could all be taken to represent a more utilitarian view rather than alignment to different ethical norms.

Therefore, simple and clear text should be used exposing the situation and lending clarity to options and consequences. Furthermore, the design challenge is in that the presented choices need to be directly associated with the ethical norm while at the same time they should provide a meaningful outcome for the story.

Designing Ethical Games

Games, like other entertainment products, use ethical dilemmas and break moral codes as a way to enhance user experience. Due to interactivity and role play, players are confronted by dilemmas embedded within the games, which exposes players to situations like corruption, theft, violence or gender discrimination just to name a few morally debatable game experiences, but, can also serve as a canvas to explore loyalty, companionship and compassion (Zagal 2009; Sicart 2009; Hartmann and Vorderer 2010; Schulzke 2009).

Rather than just a medium for conveying negative influences, a game can have a positive contribution to the ethical development of individuals and companies. In this sense, our research also opens up the opportunity to use moral judgment as a game metric. By extension, game developers can use player models (Yannakakis et al. 2013) to measure players' moral judgment in real time and alter the game to increase player engagement and enhance their play experience. This research directly contributes to player profiling by uncovering a mechanic that can be easily replicated in other interactive stories.

Opportunities

There is a clear opportunity for applying player profiling techniques to other applied fields. As described in the introduction, there are multiple domains in society, business, and healthcare whether measuring individual ethical orientation be useful. For example, there are specific studies in the financial and business sectors which show that directly ethical attitude/orientation is important to maintain ethical behaviour (Silva, Opatha, and Gamage 2018) and professional commitment (Uyar and Özer 2011), and recommend to include the ethical profiling as part of the Human Resource Management.

Despite the study being very promising and our conscious effort to recruit participants with diverse demographics (including gender, age and political affiliation) the generalizability of our results for different populations and cultural contexts would still need to be established by a replication study that involves a larger and more varied sample of participants. In addition, MES has two dimensions which have not been covered in the present study (Egoism and Relativism). For completeness future studies should address those two norms as well.

Although it is out of the scope of this study, working with game telemetry and ethics raises itself salient ethical questions about profiling players, especially for commercial purposes. There can be great societal value derived from profiling ethical orientation, which in turn calls for deliberating on ethics of player modelling as such, and for taking measures to inform and safeguard the public (e.g., to make people aware of profiling taking place and its purpose, to decide when it is morally justifiable to carry out such profiling with or without player cooperation). Guidelines for the ethical use of player modelling, informed consent procedures, and safeguards for the privacy of players and compliance with the evolving privacy legislation still need to be developed.

Conclusions

In this work, we have examined whether one's choices in an interactive story are correlated with one's moral orientation; distinguishing between Moral Equity, Contractualism and Utilitarianism. For this purpose we developed a branching interactive story, inspired by a science fiction movie, which presents players with moral dilemmas. We compared player choice behaviour within the interactive story with player's ethical orientation as measured by a questionnaire that gauges one's ethical inclination (the Multidimensional Ethics Scale). Based on that data we then applied a genetic algorithm to calculate weights to the different story's branches. Our analysis of experimental data resulted in a weighted model of player choices which is correlated with their ethical inclination and found promising results. We can therefore conclude that the identifying one's ethical inclinations through the choices they make in interactive stories is a promising direction as an alternative to questionnaires. This work adds to a growing body of research in using games to profile individuals through observing their in-game behaviour, contributing towards the broader recognition of games as an implicit measurement tool for people's attitudes and traits.

References

Association of Certified Faud Examiners. 2016. "Report To the Nations on Occupational Fraud and Abuse." *Global Fraud Study*, 1–92. https://www.acfe.com/rttn2016/docs/2016-report-to-the-nations.pdf.

Behrend, Tara S, David J Sharek, Adam W Meade, and Eric N Wiebe. 2011. "The Viability of Crowdsourcing for Survey Research." *Behavior Research Methods* 43 (October 2015): 800–813. doi:10.3758/s13428-011-0081-0.

Bies, Robert R., Matthew F. Muldoon, Bruce G. Pollock, Steven Manuck, Gwenn Smith, and Mark E. Sale. 2006. "A Genetic Algorithm-Based, Hybrid Machine Learning Approach to Model Selection." *Journal of Pharmacokinetics and*

Pharmacodynamics 33 (2): 195–221. doi:10.1007/s10928-006-9004-6.

Buchan, H. 2014. "Reidenbach and Robin's Multidimensional Ethics Scale: Testing a Second-Order Factor Model." *Psychology Research* 4 (10): 823–34.

Carr, Craig L. 1981. "The Concept of Formal Justice." *Philosophical Studies* 39 (3): 211–26. doi:10.1007/BF00354354.

Cherrington, J. Owen, and David J. Cherrington. 1992. "A Menu of Moral Issues: One Week in the Life of The Wall Street Journal." *Journal of Business Ethics* 11 (4): 255–65. doi:10.1007/BF00872167.

Cohen, Jeffrey, Yuan Ding, Cedric Lesage, and H. Stolowy. 2010. "Corporate Fraud and Managers' Behavior: Evidence from the Press." *Journal of Business Ethics* 95 (SUPPL. 2): 271–315. doi:10.1007/s10551-011-0857-2.

Crawford, Chris. 2008. "The Art Game Design." Edited by Array. *Epilepsy Research* 49 (Clinical Trial PT-Journal Article PT-Multicenter Study PT). Elsevier: 512.

El-Nasr, Magy Seif, Anders Drachen, and Alessandro Canossa. 2013. *Game Analytics: Maximizing the Value of Player Data*. doi:10.1007/978-1-4471-4769-5.

Ewell, Patrick J., Rosanna E. Guadagno, Matthew Jones, and Robert Andrew Dunn. 2016. "Good Person or Bad Character? Personality Predictors of Morality and Ethics in Avatar Selection for Video Game Play." *Cyberpsychology, Behavior, and Social Networking* 19 (7). Mary Ann Liebert, Inc. 140 Huguenot Street, 3rd Floor New Rochelle, NY 10801 USA : 435–40. http://online.liebertpub.com/doi/10.1089/cyber.2015.0207.

Flory, Steven M, Thomas J Phillips, R. Eric Reidenbach, and Donald P. Robin. 1992. "A Multidimensional Analysis of Selected Ethical Issues in Accounting." *The Accounting Review* 67 (2): 284–302.

Gabbiadini, A., P. Riva, L. Andrighetto, C. Volpato, and Brad J. Bushman. 2014. "Interactive Effect of Moral Disengagement and Violent Video Games on Self-Control, Cheating, and Aggression." *Social Psychological and Personality Science* 5 (4): 451–58. doi:10.1177/1948550613509286.

Goldberg, David Edward. 1989. *Genetic Algorithms in Search, Optimization and Machine Learning*. *Addison-Wesley*. doi:10.5860/CHOICE.27-0936.

Gupta, Seema. 2010. “A Multidimensional Ethics Scale for Indian Managers ’ Moral Decision Making.” *Electronic Journal of Business Ethics and Organization Studies* 15 (1): 5–14. doi:10.1007/s10551-011-0910-1.

Hartmann, Tilo, and Peter Vorderer. 2010. “It’s Okay to Shoot a Character: Moral Disengagement in Violent Video Games.” *Journal of Communication* 60 (1): 94–119. doi:10.1111/j.1460-2466.2009.01459.x.

Hoffman, W. Michael, Robert E. Frederick, and Mark S. Schwartz. 2014. *Business Ethics. Readings and Cases in Corporate Morality*. Edited by W. Michael Hoffman, Robert E. Frederick, and Mark S. Schwartz. *Business Ethics. Readings and Cases in Corporate Morality*. 5 edition. Vol. 1. Wiley-Blackwell. doi:10.1017/CBO9781107415324.004.

Humphreys, Neil, Donald P Robin, R Eric Reidenbach, and Donald L Moak. 1993. “The Ethical Decision Making Process of Small Business Owner/Managers and Their Customers.” *Journal of Small Business Management* 31 (3): 9–22.

Imanishi-kari, Thereza, Scientific Integrity, and Murray Hill. 2006. “Ethics and Fraud.” *Nature* 439 (7073): 117–18. doi:10.1038/439117a.

Jung, Insung. 2009. “Ethical Judgments and Behaviors: Applying a Multidimensional Ethics Scale to Measuring ICT Ethics of College Students.” *Computers and Education* 53 (3): 940–49. doi:10.1016/j.compedu.2009.05.011.

Kant, Immanuel. 2000. “Foundations of the Metaphysics of Morals.” *World Philosophers & Their Works JN - World Philosophers & Their Works*, 1–5. doi:10.1016/B978-0-12-373932-2.00210-6.

Kohlberg, L. 1981. “Essays in Moral Development.” In *The Philosophy of Moral Development*.

KPMG. 2013. “KPMG Malaysia Fraud, Bribery and Corruption Survey 2013.” *KPMG Forensic*, 1–72. doi:10.1007/s13398-014-0173-7.2.

Lankveld, Giel Van, Sonny Schreurs, and Pieter Spronck. 2009. “Psychologically Verified Player Modelling.” In *GAMEON*, 12–19. http://www.gielvanlankveld.nl/literature/Psychologically Verifed Player Modelling.pdf.

Lankveld, Giel van, Pieter Spronck, Jaap van den Herik, and Arnoud Arntz. 2011.

"Games as Personality Profiling Tools." *2011 IEEE Conference on Computational Intelligence and Games (CIG'11)*. IEEE, 197–202. doi:10.1109/CIG.2011.6032007.

Lloyd, Liz. 2004. "Mortality and Morality: Ageing and the Ethics of Care." *Ageing and Society*. doi:10.1017/S0144686X03001648.

Longenecker, Justin G., Joseph A. McKinney, and Carlos W. Moore. 1988. "Egoism and Independence: Entrepreneurial Ethics." *Organizational Dynamics* 16 (3): 64–72. doi:10.1016/0090-2616(88)90037-X.

McDevitt, Roselie, Catherine Giapponi, and Cheryl Tromley. 2007. "A Model of Ethical Decision Making: The Integration of Process and Content." *Journal of Business Ethics* 73 (2): 219–29. doi:10.1007/s10551-006-9202-6.

Mulgan, Tim. 2007. *Understanding Utilitarianism*. *Classical Utilitarianism*. doi:10.1017/UPO9781844653904.

Nacke, Lennart E, Anders Drachen, Kai Kuikkaniemi, and Yvonne a W De Kort. 2009. "Playability and Player Experience Research." *Proceedings of the IEEE*, 1–11. http://www.bth.se/fou/forskinfo.nsf/17e96a0dab8ab6a1c1257457004d59ab/e0a8cdd8cfc0c7e6c125762c005557c0!OpenDocument.

Newman, Edward, and M.C. McFarland. 1990. "Comments, with Reply, on 'Urgency of Ethical Standards Intensifies in Computer Community' (Standards)." *Computer* 23 (6): 6. doi:10.1109/2.55495.

Pereira Santos, Carlos, Kevin Hutchinson, Vassilis-Javed Khan, and Panos Markopoulos. 2017. "Measuring Self-Esteem with Games." In *ACM Intelligent User Interfaces*. ACM Intelligent User Interfaces: ACM. doi:http://dx.doi.org/10.1145/3025171.3025196.

Pereira Santos, Carlos, Vassilis-Javed Khan, and Panos Markopoulos. 2016. "Inferring A Player's Need For Cognition From Hints." In *Proceedings of the 21st International Conference on Intelligent User Interfaces*, 76–79. IUI '16. New York, NY, USA: ACM. doi:10.1145/2856767.2856805.

Rand, Ayn. 1964. *The Virtue of Selfishness: A New Concept of Egoism*. *Signet, New York, NY*. New American Library.

Randall, D. M., and A. M. Gibson. 1990. "Methodology in Business Ethics Research: A

Review and Critical Assessment." *Journal of Business Ethics* 9 (6): 457–71. doi:10.1007/BF00382838.

Rawls, John. 1999. *A Theory of Justice*. *Harvard University Press*. Harvard University Press - Belknap.

Reidenbach, R. E., and D. P. Robin. 1990. "Toward the Development of a Multidimensional Scale for Improving Evaluations of Business Ethics." *Journal of Business Ethics* 9 (8): 639–53. doi:10.1007/BF00383391.

Reidenbach, R. Eric, Donald P. Robin, and Lyndon Dawson. 1991. "An Application and Extension of a Multidimensional Ethics Scale to Selected Marketing Practices and Marketing Groups." *Journal of the Academy of Marketing Science* 19 (2): 83–92. doi:10.1007/BF02726000.

Scanlon, Thomas M. 1982. "Contractualism and Utilitarianism." In *Utilitarianism and Beyond*, edited by Amartya Kumar Sen and Bernard Arthur Owen Williams, 103–10. Cambridge University Press.

Schulzke, Marcus. 2009. "Moral Decision Making in Fallout." *Game Studies* 9 (2).

Sicart, Miguel. 2009. "The Banality of Simulated Evil: Designing Ethical Gameplay." *Ethics and Information Technology* 11 (3): 191–202. doi:10.1007/s10676-009-9199-5.

Silva, Viruli, H Opatha, and Aruna Gamage. 2018. "Does Ethical Orientation of HRM Impact on Employee Ethical Attitude and Does Ethical Orientation of HRM Impact on Employee Ethical Attitude and Behavior ? Evidence from Sri Lankan Commercial Banks." *International Business Research* 11 (1). doi:10.5539/ibr.v11n1p218.

Soltani, Bahram. 2014. "The Anatomy of Corporate Fraud: A Comparative Analysis of High Profile American and European Corporate Scandals." *Journal of Business Ethics* 120 (2): 251–74. doi:10.1007/s10551-013-1660-z.

Uyar, Metin, and Gökhan Özer. 2011. "The Ethical Orientation and Professional Commitment : An Empirical Examination on Turkish Accountants" 5 (23): 10023–37. doi:10.5897/AJBM10.1521.

Williamson, Stan, Kenneth E. Clow, Bruce C. Walker, and T. Selwyn Ellis. 2011. "Ethical Issues in the Age of the Internet: A Study of Students' Perceptions Using

the Multidimensional Ethics Scale." *Journal of Internet Commerce* 10 (February 2015): 128–43. doi:10.1080/15332861.2011.571992.

Worth, Narnia C., and Angela S. Book. 2014. "Personality and Behavior in a Massively Multiplayer Online Role-Playing Game." *Computers in Human Behavior* 38. Elsevier Ltd: 322–30.

Yannakakis, Georgios N., Pieter Spronck, Daniele Loiacono, and Elisabeth André. 2013. "Player Modeling." *Dagstuhl Follow-Ups*. Schloss Dagstuhl - Leibniz-Zentrum fuer Informatik. doi:10.4230/DFU.Vol6.12191.45.

Yee, Nick, Nicolas Ducheneaut, Les Nelson, and Peter Likarish. 2011. "Introverted Elves & Conscientious Gnomes: The Expression of Personality in World of Warcraft." *Proceedings of the Annual Conference on Human Factors in Computing Systems - CHI '11*, 753–62.

Zagal, José P. 2009. "Ethically Notable Videogames: Moral Dilemmas and Gameplay." *Proceedings of DiGRA 2009*, 9.